

Scores and Behavioral Evidence in Agent Reinforcement Learning

From Task Scores to Agent Behavior

A behavioral study of Qwen3-8B on Ask-or-Act and WebShop.

Which behaviors improve when the training score rises? We compare SFT, GRPO after SFT, and direct GRPO on Ask-or-Act, and examine visit rewards and within-group training signals on WebShop.

Action Order on Original and Modified Tasks

We first train with SFT on 192 demonstration trajectories, then continue with GRPO. Success on the original email training tasks rises from 178/192 to 192/192. Both models complete all 20 clarification–confirmation paired tasks, while the count of tasks with premature action falls from 19/20 to 0/20. Here, premature action means attempting a write before the task’s action conditions are met, including attempts blocked by the environment. This set contains ten controls and ten tasks combining the requirements.

The direction reverses on another set of email tasks. Original file queries return sharing rules directly. The new tasks distribute file size and sharing thresholds across two tools, requiring retrieval before the model decides whether to send. Both checkpoints succeed on the same 50 of 60 tasks, but the count satisfying the pre-write lookup condition falls from **60/60 to 0/60**. The SFT pre-write lookup count includes ten tasks that end after lookup without a write attempt.

Here is the recorded order on one successful task, omitting the shared initial queries. The file is 4 MB, the approval threshold is 10 MB, and the sharing prohibition limit is 25 MB.

SFT

Query sharing policy → request confirmation → send succeeds → finish

SFT followed by GRPO

Attempt to send, blocked by the environment: missing sharing rules

→ query sharing policy → request confirmation → send succeeds → finish

Both models also request confirmation unnecessarily on this task. The same training run reduces premature action on the original paired tasks but does not preserve pre-write lookup on the two-tool tasks.

Behavioral Differences Across Three Training Routes

The report studies SFT, GRPO after SFT, and GRPO directly from the initial Instruct model. Direct GRPO satisfies the pre-write lookup condition on 58/60 of the two-tool tasks, yet all tasks still contain premature action; in 28, the flag is triggered solely by an operation that does not match the request—for example, attempting to create a draft when the user asked to send an email.

Scheduling reveals another distinction. A group query returns an owner identifier that the model should pass to a member query. Direct GRPO completes all 30 tasks requiring this dependency but actually passes the returned identifier in only six. One successful trajectory first misuses a group identifier, receives a blocked call, then obtains the owner's member identifier from the user and completes the booking.

Tied Returns and Repeated Actions

After one WebShop GRPO run, we sample eight trajectories at temperature 1.0 on each of 500 tasks. On 197 tasks, all eight receive the same score between 0 and 1. There is room to improve, but standard GRPO's group-relative reward advantage is zero. On **121 of these tasks, all eight environment-action sequences are also identical.**

The remaining 76 intermediate tied tasks contain different actions. For these tasks, first determine whether the difference matters to the requirements, then check whether the reward assigns different scores to those task-relevant behaviors. Tasks with the same behavior in all eight samples need different candidates; rescoreing alone does not create a new behavior. Once rewards differ, the action advantage actually used for the update shows how scoring enters training.

WebShop's visit reward increases subpage visitation before purchase from 94/500 to 498/499. The full report also compares purchase fields and action sequences across page-text treatments and locates target attributes in the products actually purchased.

Applications to Evaluation and Training Diagnosis

The project provides a public implementation of task generation, tool environments, and deterministic evaluation.

The full report retains positive results and failed experiments across the three training routes. Appendix D.1 describes the public implementation and statistical rules. Readers can apply these methods to compare checkpoints, design behavioral regression tests, and determine whether a training anomaly involves evaluation, sampling, or advantage computation.

Continue to the full English technical report →